

User Manual

How to use CodeSage AI, step by step.

CodeSage checks Java projects on GitHub for technical debt. It scans the code, gives each project a health score from 0 to 100, and ranks the problems worth fixing first, so a team can spend limited refactoring time where it matters most. This manual walks through the product in the order you will use it, from creating an account to reading results and managing a team.

Website	codesageai.dev
Version	Release 1.0, October 2026
Works with	Public GitHub repositories that contain Java source code
You need	A laptop or desktop browser (Chrome, Edge or Firefox) and an email address you can open

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01 CodeSage at a glance

CodeSage reads the Java code in a repository, finds places where the code will be harder to change later (technical debt), and turns them into a ranked to-do list. It never changes your repository; it only reads it.

Words used in this manual

Term	What it means
Workspace	Your team's private area. It holds projects, scans, scoring profiles and teammates. Nothing is shared between workspaces.
Project	A GitHub repository connected to a workspace.
Scan	One analysis of one branch at one commit. A successful scan saves a snapshot .
Finding	One piece of detected technical debt at a file and line, with a severity (critical, high, medium or low).
Health score and grade	A score from 0 to 100 and a letter from A (best) to E. Higher means easier to maintain.
Refactor first	The ranked list of findings. Items at the top are the best value to fix first.
Scoring profile	A set of weights that decides how much each kind of debt counts toward the score.
Role	What you are allowed to do in a workspace: org admin, manager, developer or viewer.

02 Create your account and sign in

CodeSage uses Asgardeo, a secure sign-in service, for accounts. You sign in on the Asgardeo page and come straight back; CodeSage never sees your password.

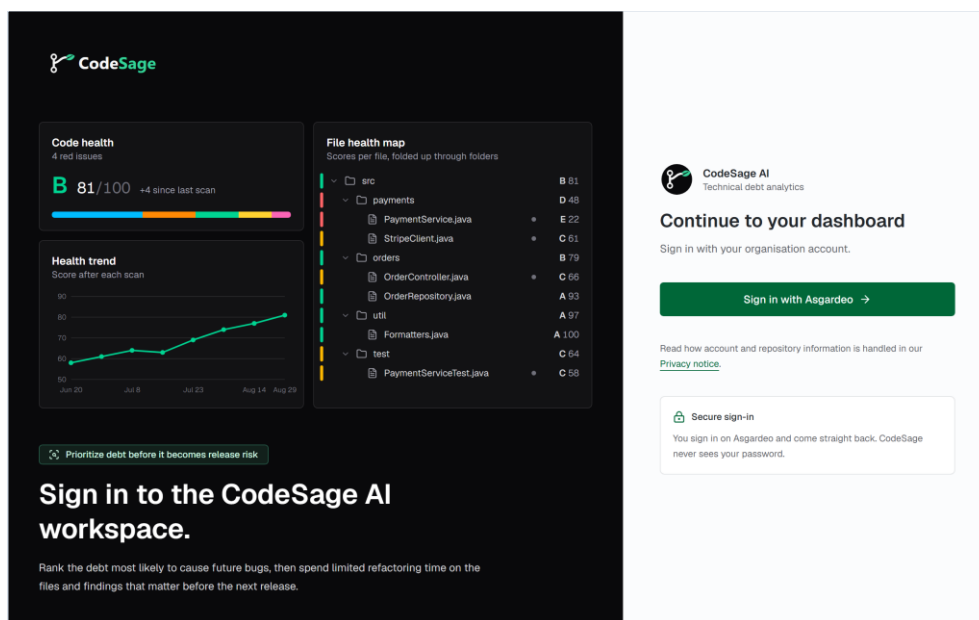


Figure 1 The CodeSage sign-in page at codesageai.dev.

FIRST TIME

- Open the website**
Go to codesageai.dev and select **Sign in with Asgardeo**.

You will see
The Asgardeo Sign In page.

- | | |
|--|---|
| 02 Register
Select Register . Enter your email address and a password of 8 to 64 characters with at least one number, one uppercase and one lowercase letter. Select Sign Up . | You will see
A message asking you to confirm your email. |
| 03 Confirm your email
Open the email from Asgardeo and follow the confirmation link. | You will see
Your account is active. |
| 04 Sign in
Back on the Sign In page, type your email address as the username , enter your password and select Sign In . | You will see
The CodeSage Workspace page. |

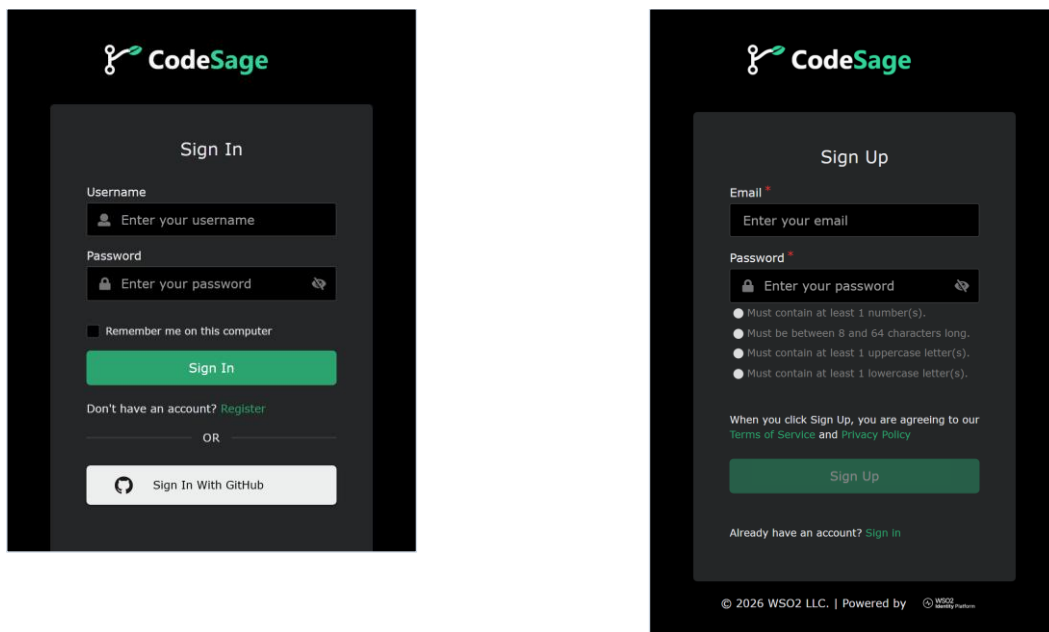


Figure 2 Asgardeo Sign In page (left) and Register page (right).

Good to know

On your first sign-in CodeSage creates a workspace called **My Workspace** and makes you its **org admin**. If someone invited you to their workspace, you will also see that workspace in the workspace picker (see section 4).

03 Take the guided trial

The first time you sign in, a short guided trial opens by itself. Each card points at one part of the screen and explains it. Use **Next** and **Back** to move, **Skip** to jump past a part you already understand, or the **x** to close the trial.

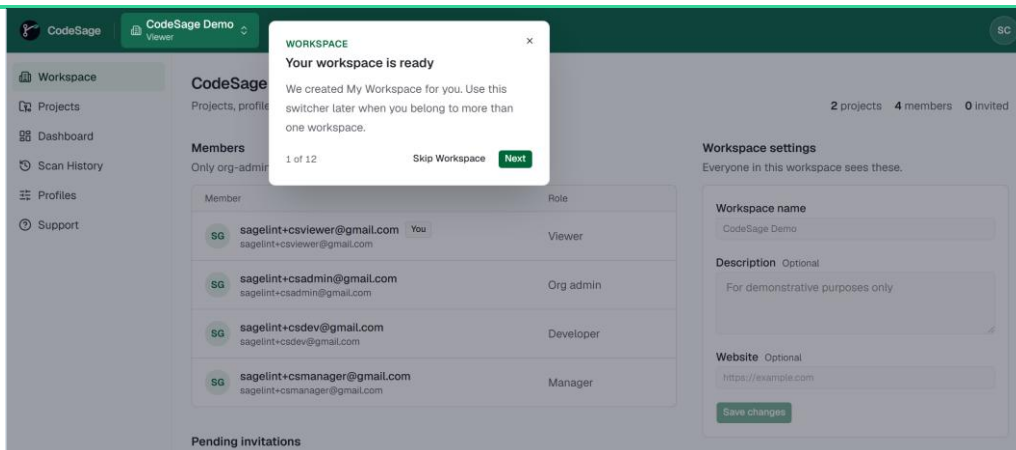


Figure 3 The first step of the guided trial, pointing at the workspace picker.

You can replay the trial at any time. Open **Support** in the side menu and choose **Prefer a walkthrough? Take the New User Trial**, or go to the New User Trial page directly. From there select **Start full trial**, or pick one lesson (Workspace, Projects, Dashboard, Profiles or Scan History) to learn only that part.

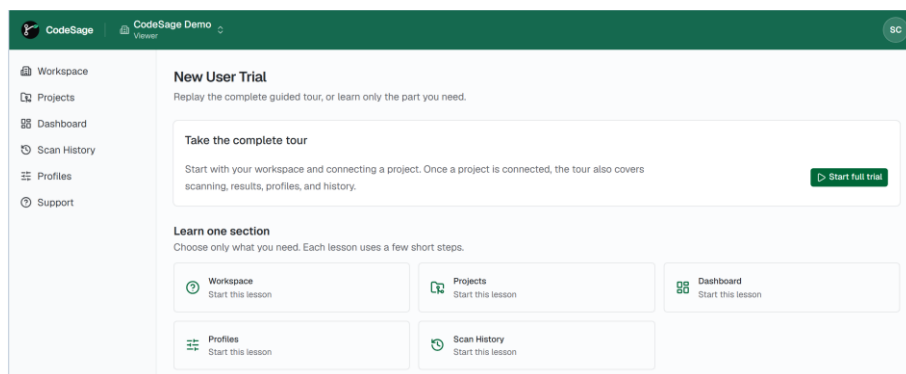


Figure 4 The New User Trial page: replay the full trial or a single lesson.

04 Find your way around

Every page has the same layout: a green top bar, a side menu on the left, and the page content in the middle.

Side menu

Item	What it opens
Workspace	Workspace details, team members, invitations and settings.
Projects	Connected repositories, and the box to connect a new one.
Dashboard	Health, findings and file map for the selected project.
Scan History	Every saved scan of the selected project.
Profiles	Scoring profiles and which one each project uses.
Support	The help centre, guides and the New User Trial.
Theme	Light, dark or system colour scheme.
Sign out...	Ends your session (asks you to confirm first).
Collapse sidebar	Shrinks the side menu to icons to give the page more room.

Top bar

The top bar always shows where you are. From left to right: the **workspace picker** (with your role under the name), the **project picker**, the **branch**, the commit of the result you are looking at, the profile used for scoring, the scan navigator (**Latest 1/1**), the **Scan** button and your account badge. The project and branch parts appear once you open a project.

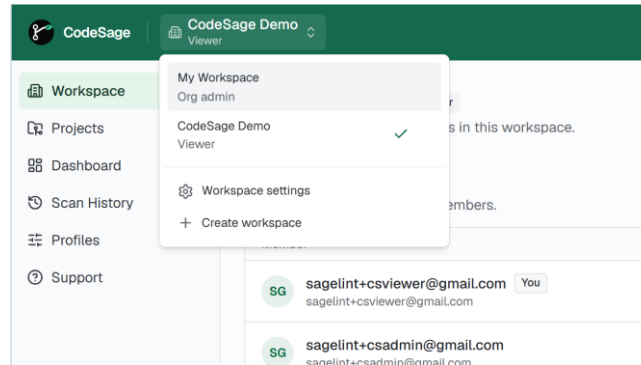


Figure 5 The workspace picker. A tick marks the active workspace; your role shows under each name.

Tip

If a button looks greyed out, your role in the active workspace does not allow that action. Section 5 lists what each role can do.

05 Your workspace and team

Open **Workspace** in the side menu. Org admins see the full page below; other roles see the member list and settings as read-only.

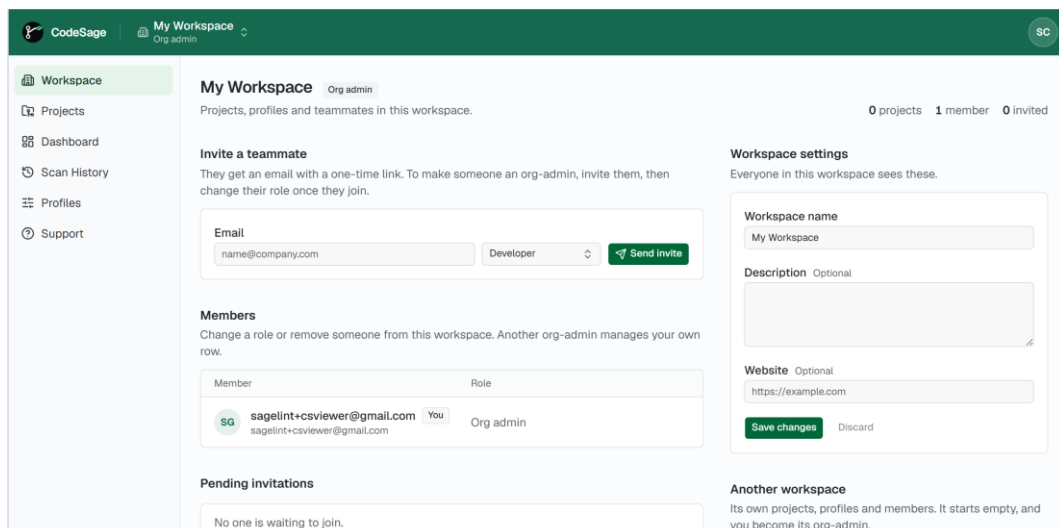


Figure 6 The Workspace page for an org admin, with the invite form, members and settings.

Change workspace details

Under **Workspace settings**, edit the **Workspace name**, an optional **Description** and an optional **Website**, then select **Save changes**. The new name appears in the workspace picker straight away.

Invite a teammate

01	Enter their email Under Invite a teammate , type the person's email address.	You will see The address in the Email box.
02	Choose a role Pick Org admin , Manager , Developer or Viewer from the list.	You will see The role next to the address.
03	Send the invite Select Send invite .	You will see The person under Pending invitations .
04	They accept The teammate opens the one-time link in the email and signs in with the same email address that was invited.	You will see They appear under Members with that role.

From the same page an org admin can revoke a pending invitation, change a member's role, or remove a member. **Another workspace** → **New workspace** creates a fresh, empty workspace where you become org admin. **Danger zone** permanently deletes the workspace and everything in it (your GitHub repositories are not affected).

What each role can do

Action	Org admin	Manager	Developer	Viewer
View projects, dashboards, findings, history and profiles	✓	✓	✓	✓
Start a scan, and stop a scan you started	✓	✓	✓	–
Mark findings done or reopen them	✓	✓	✓	–
Connect or remove repositories	✓	✓	–	–
Stop anyone's scan	✓	✓	–	–
Change scoring profiles and the workspace default	✓	✓	–	–
Invite or remove members and change roles	✓	–	–	–
Edit workspace settings or delete the workspace	✓	–	–	–

06 Connect a repository

A project is a link between your workspace and a public GitHub repository that contains Java code. Connecting needs the **manager** or **org admin** role.

01	Open Projects Select Projects in the side menu.	You will see The Connect a GitHub repository box.
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02 Paste the URL

Paste the full repository address, for example
`https://github.com/JodaOrg/joda-time`.

You will see
The address in the box.

03 Connect

Select **Connect repository**.

You will see
The repository under **Connected repositories**.

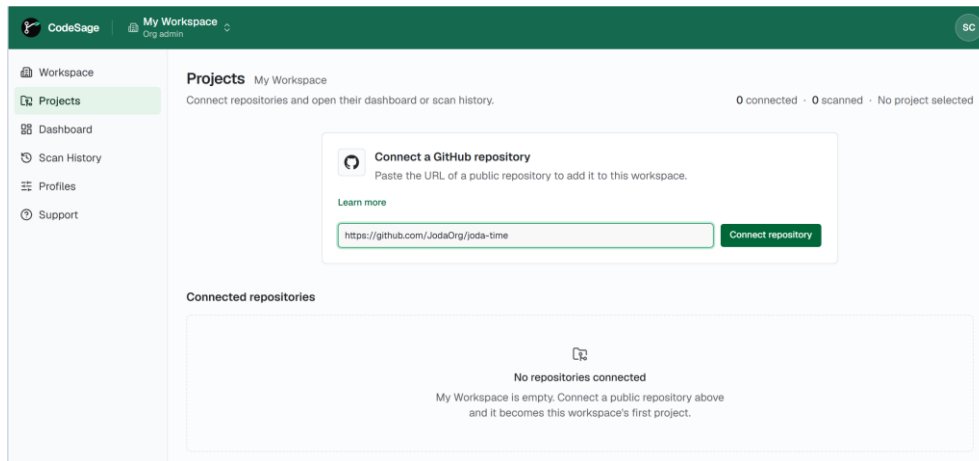


Figure 7 Connecting the sample repository JodaOrg/joda-time to an empty workspace.

Each connected repository shows its latest health score and grade, the branch it was scanned on and when it was connected. Use the buttons on its row:

- **Select** makes it the active project, so **Dashboard** and **Scan History** in the side menu open it.
- **Go to dashboard** opens its results.
- **History** opens its saved scans.
- The arrow icon opens the repository on GitHub.

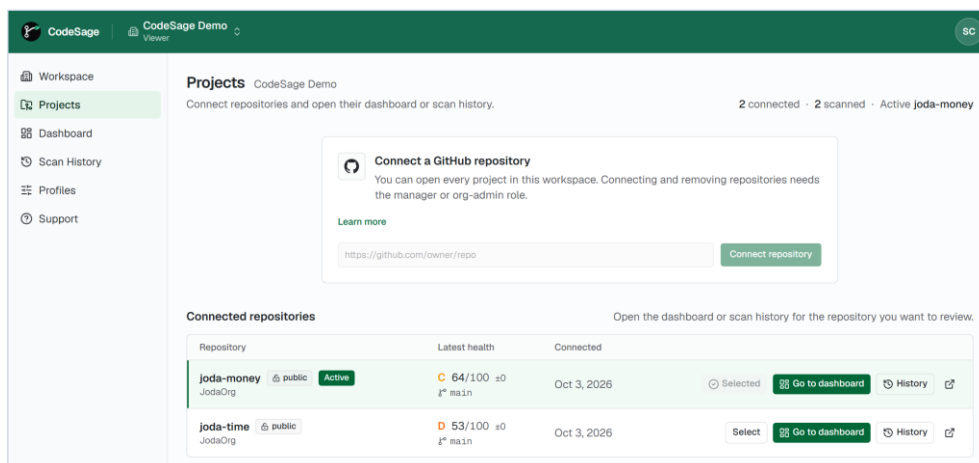


Figure 8 Two connected repositories with their latest grades.

Note

Removing a project deletes its CodeSage scans and findings from this workspace only. It never changes the repository on GitHub. Private repositories and other languages are not supported in this release; a branch without Java files is rejected with a **No Java files** message.

07 Run a scan

01	Open the dashboard On Projects , select Go to dashboard for the repository.	You will see The project dashboard.
02	Choose a branch Use the branch picker in the top bar. The default branch (usually main) is selected for you.	You will see The branch name in the top bar.
03	Start the scan Select Scan at the top right. Developers, managers and org admins can scan.	You will see A progress panel showing each stage.
04	Wait for the result You can keep using CodeSage; progress follows you across pages. When it finishes, select Show them to open the new result.	You will see The dashboard with the new scores.



Figure 9 The top bar on a dashboard: workspace, project, branch, commit, profile, scan navigator and the Scan button.

During a scan CodeSage copies the repository, measures the Java files and their Git history, detects findings, and calculates the report. A large project can take several minutes. **Queued** means another scan in the workspace is running and yours will start next. You can stop a scan; it stops between stages, so your last complete result is never replaced by a half-finished one.

08 Read the dashboard

The dashboard opens in the **Overview** view, which shows four panels at once.

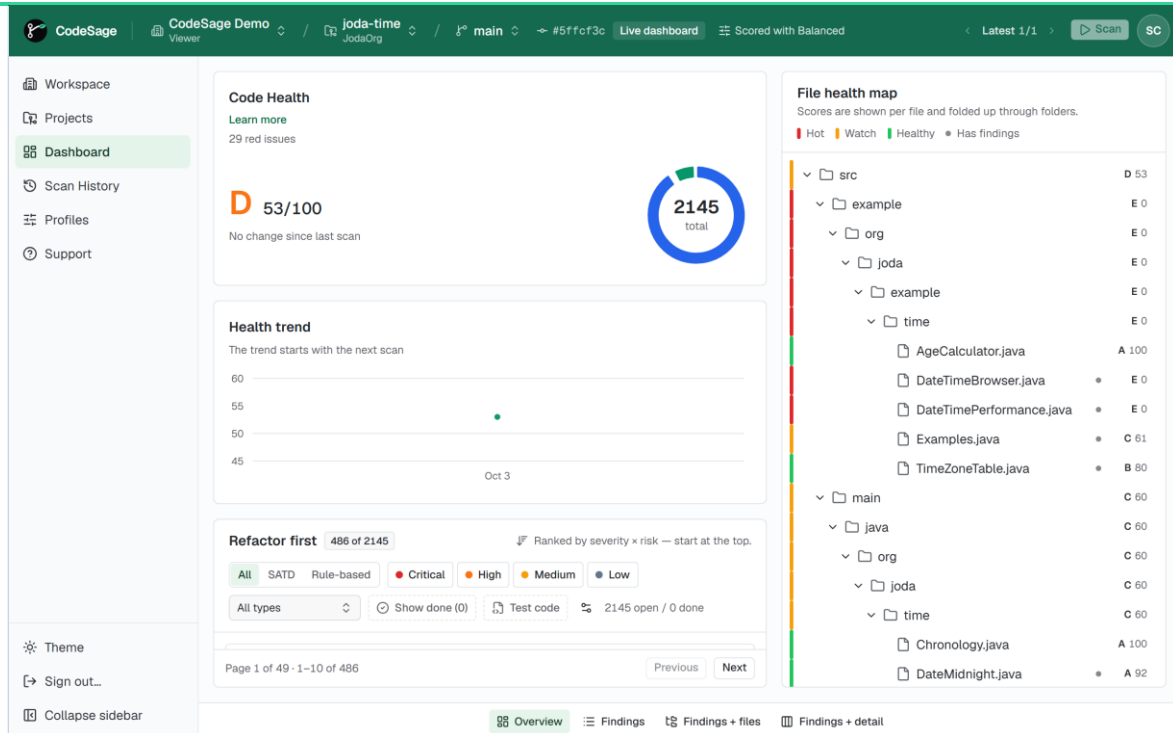


Figure 10 Dashboard overview for JodaOrg/joda-time.

Panel	How to read it
Code Health	The grade letter and score out of 100, how many red (critical and high) issues there are, the total number of findings in the ring, and the change since the last scan.
Health trend	The score after each scan of this branch. The line starts once there are two scans.
Refactor first	All findings, ranked so the best value to fix is at the top (see section 9).
File health map	Every folder and file with its own grade. The coloured bar on the left shows Hot (red), Watch (orange) or Healthy (green); a dot means the file has findings. Folder scores are folded up from the files inside them.

What the grade means

Grade	A	B	C	D	E
Score	85 or more	70 to 84	55 to 69	40 to 54	below 40

The score is a guide to how maintainable the code is, measured against the amount of Java code. It is not a test-coverage or correctness score.

09 Work with findings

Each finding card shows its severity, its category (for example **code-design**), where it came from (**rule** for static analysis, **SATD** for self-admitted technical debt, meaning comments in which developers admit a shortcut, found by CodeSage's machine-learning model), the file and line, and a priority such as **P7**. Severity says how serious the issue is; priority decides its place in the list.

Filter the list

- **All**, **SATD** and **Rule-based** choose where findings came from.
- **Critical**, **High**, **Medium** and **Low** each switch one severity on or off. To see only high findings, switch off the other three.
- **All types** narrows the list to one category, such as security or documentation.
- **Show done** brings back findings marked done; **Test code** includes findings in test files.
- Use **Previous** and **Next** at the bottom to page through the list.

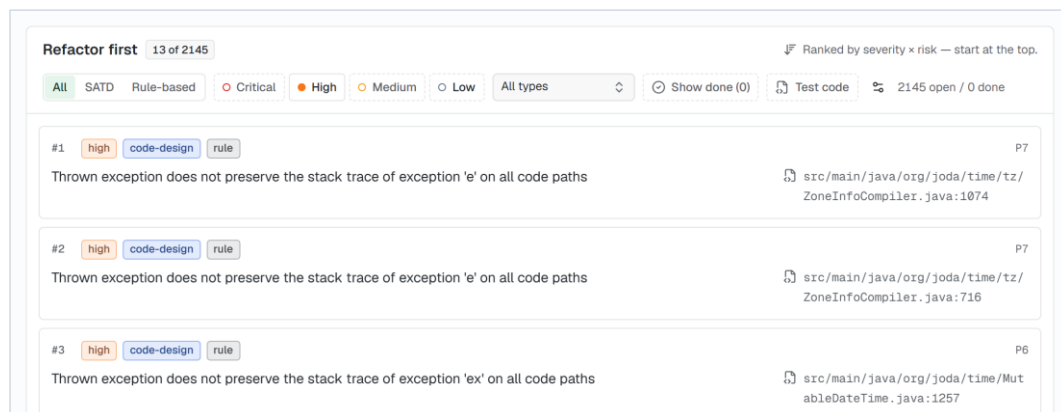


Figure 11 Refactor first filtered to high-severity findings only (13 of 2145).

Choose a view

The bar at the bottom of the dashboard switches between four views. Your browser remembers your choice.

View	Best for
Overview	Health, trend, findings and file map together.
Findings	A wide findings list for working through priorities.
Findings + files	The list beside the file tree, to see where findings cluster.
Findings + detail	The list beside the full detail of the selected finding.

Open a finding

In **Findings + detail**, select any card. The detail panel shows the file and line, **What PMD found** (for rule-based findings), **Why this matters** and **How to fix it**, with a link to the rule documentation where one exists.

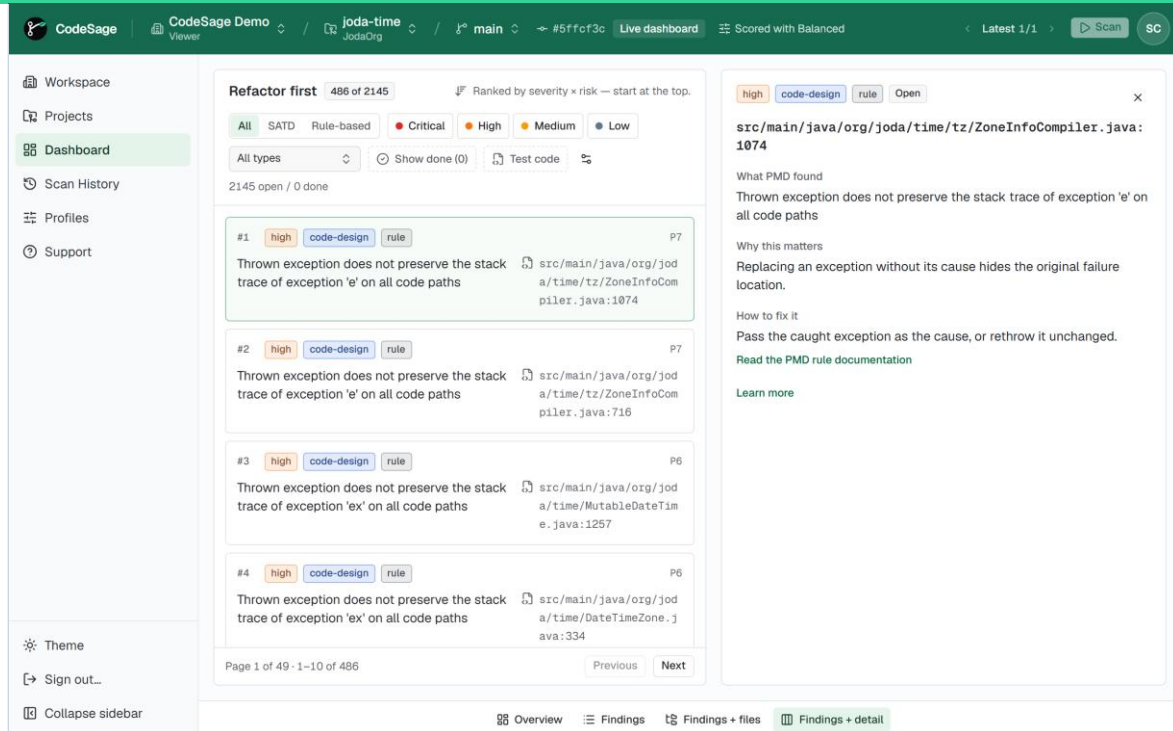


Figure 12 Findings + detail view with the top-ranked finding open.

Done

Developers and above can mark a finding **done** when the work is planned or finished. Done hides it from the open list but does not change the code or the score. Fix the code, then scan again to see the score improve.

10 Scoring profiles

A profile decides how much each kind of debt counts toward the health score. Changing a profile re-scores current and past results immediately, with no new scan. Profiles change scoring only, never what is detected. Changing them needs the **manager** or **org admin** role.

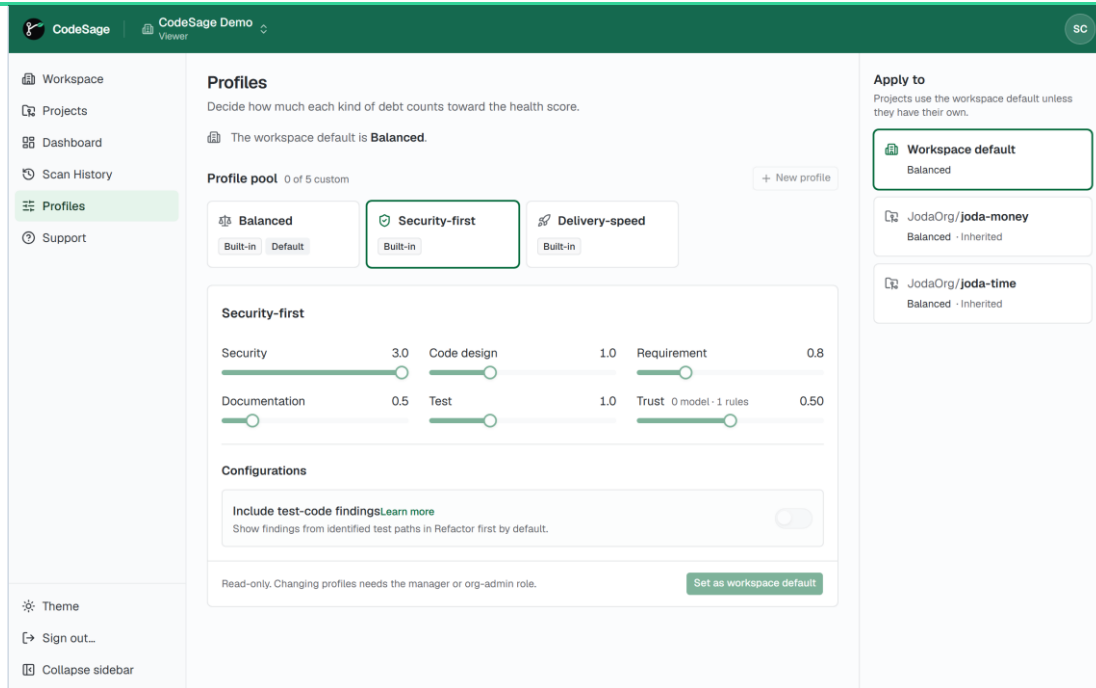


Figure 13 The Profiles page with the built-in Security-first profile selected.

- **Built-in profiles:** **Balanced** (everything counts equally), **Security-first** (security weighs 3.0) and **Delivery-speed**. They cannot be edited.
- **Category weights** run from 0.1 to 3.0 for Security, Code design, Requirement, Documentation and Test.
- **Trust** sets how much CodeSage relies on the machine-learning model (0) versus the analysis rules (1).
- **Include test-code findings** decides whether findings in test files affect the score.
- **Set as workspace default** makes the selected profile apply to every project.
- The **Apply to** panel on the right gives one project its own profile, overriding the default.

To make your own, select **New profile**, give it a name, set the sliders and save. A workspace can keep five custom profiles.

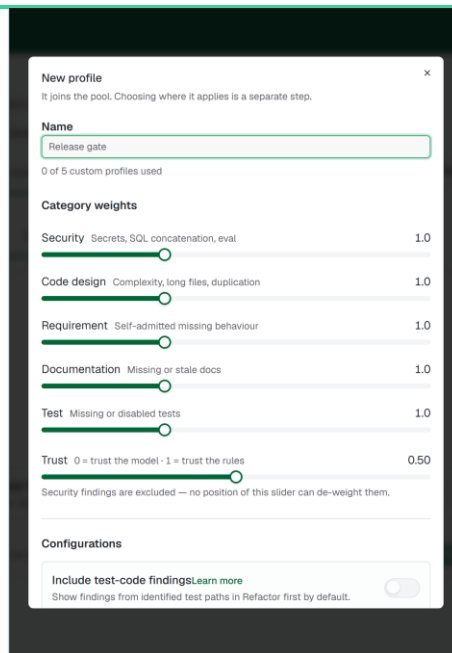


Figure 14 Creating a custom profile.

11 Scan history

Every successful scan is kept as a snapshot. Open **Scan History** in the side menu to see them, newest first, with the branch, commit, score, grade and the change from the previous scan.

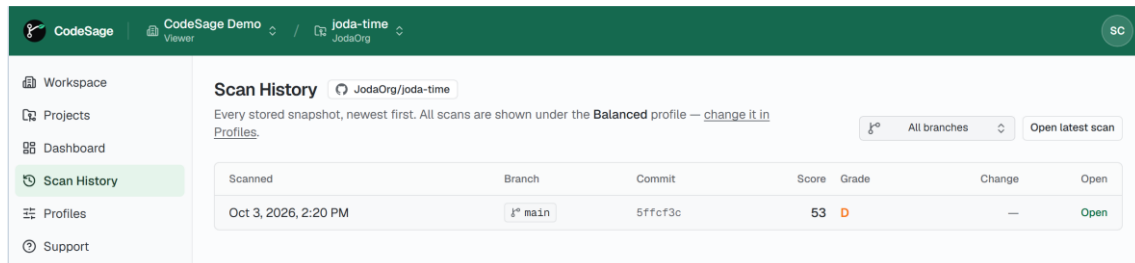


Figure 15 Scan History for JodaOrg/joda-time.

- Use **All branches** to show one branch only.
- Select **Open** on a row to view the dashboard exactly as that scan found it.
- Select **Open latest scan** (or use the arrows beside **Latest** in the top bar) to return to the newest result.
- The files and findings in a snapshot never change. The score shown uses the profile in force today, so it can change after a profile edit.

12 Help and support

Select **Support** in the side menu to open the help centre. Type a question into the search box, or browse topics such as **How analysis works**, **Data and privacy**, the dashboard, findings, profiles and troubleshooting.

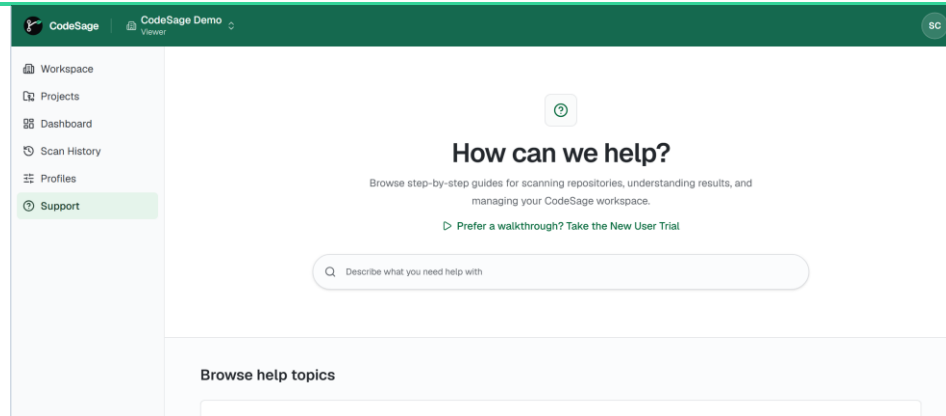


Figure 16 The help centre.

The **Product Guide** at codesageai.dev/guide summarises every area on one page, including the severity guide and the ranges for scores and weights. It is public, so you can share it with someone who does not have an account yet.

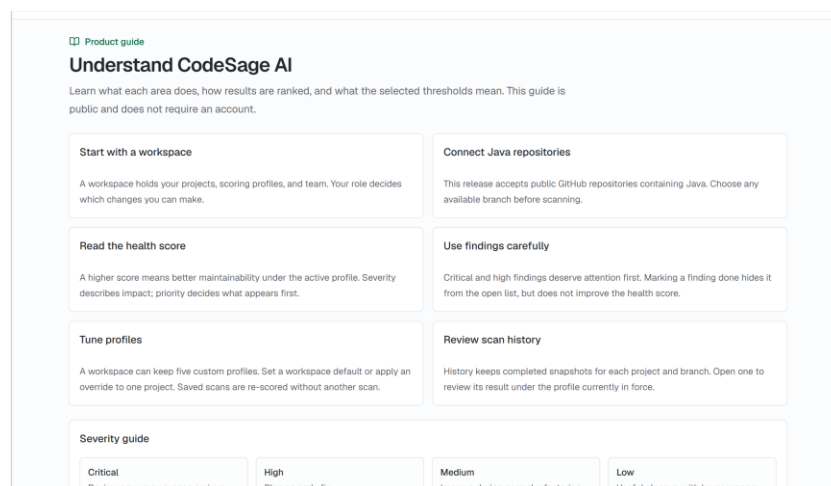


Figure 17 The public Product Guide.

13 Theme and signing out

Select **Theme** at the bottom of the side menu and choose **Light**, **Dark** or **System default** (follow your computer's setting). To leave, select **Sign out...** and confirm with **Sign out**. You will sign in through Asgardeo again next time.

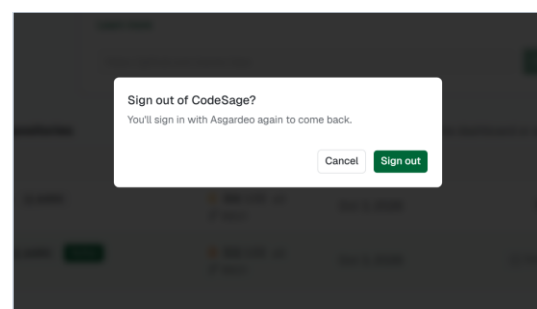
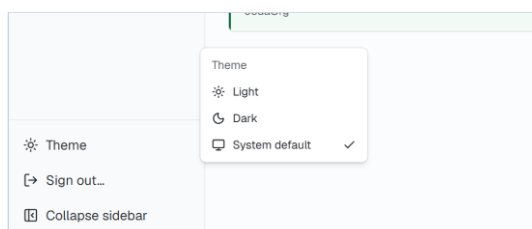


Figure 18 The Theme menu (left) and the sign-out confirmation (right).

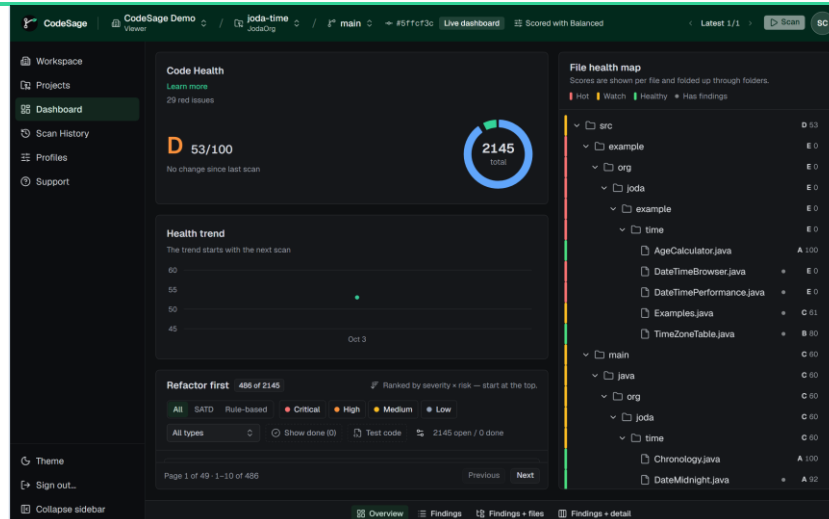


Figure 19 The dashboard in dark mode.

14 Troubleshooting

Problem	What to do
I cannot sign in	Use your email address as the username . If you just registered, confirm your email first. Use the password reset on the Asgardeo page if you have forgotten it.
A button is greyed out	Your role in the active workspace does not allow it. Check the role under the workspace name in the top bar and ask an org admin for a different role.
My repository is rejected	Check that it is a public GitHub repository and that the branch contains Java files. Paste the full URL, starting with https://github.com/ .
My scan stays queued	Another scan in the workspace is running; yours starts when it finishes. Do not start the same scan again.
The score changed without a scan	Someone changed the scoring profile. Scores are recalculated from the saved findings when a profile changes.
I do not see my teammate's workspace	Open the invitation email and sign in with the exact email address that was invited, then use the workspace picker.
A page does not load	Use Retry or Try again if offered, check the active workspace and project, and refresh once. If it keeps failing, contact us with the time and the exact message.

Privacy

When reporting a problem, never send passwords, session cookies or access tokens.

Something not working?

If you cannot sign in or the site shows an error, contact us:

Janidu Pabasara	janidup.23@cse.mrt.ac.lk
Chamodh Nethsara	chamodhn.23@cse.mrt.ac.lk
Nathasha Nethmini	nathashan.23@cse.mrt.ac.lk

Thank you for using CodeSage.